Advent Tree

Mechanical advent calendar · Assembly manual for the main model



24 little boxes hang on a tree of four tiers. A single drum runs through the centre of the tree; a knob on top turns it one click per day. Every box is held by a small latch and a pair of magnets — only the box for the day the knob points at can be pulled out. This manual takes you through the assembly from the base upwards.

This manual covers the knob version only. The motor kit is not part of it.

What is in the box



Printed parts

Base with the bottom tray and tube (tier 4) — 1

Tier carriers (tray + tube), tiers 3, 2, 1 — 3

Drum segments, one per tier — 4

Latches — 24

Boxes, two halves each (A and B, numbers inlaid) — 24

Top floor with the dial — 1

Knob — 1

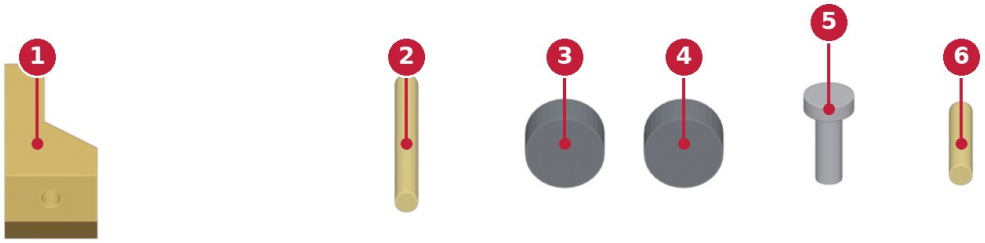
Cap (top of the tree) — 1

Star halves — 2

Tools

Super glue (CA), flush cutters, tweezers, a file or fine sandpaper, a felt pen to mark magnet poles

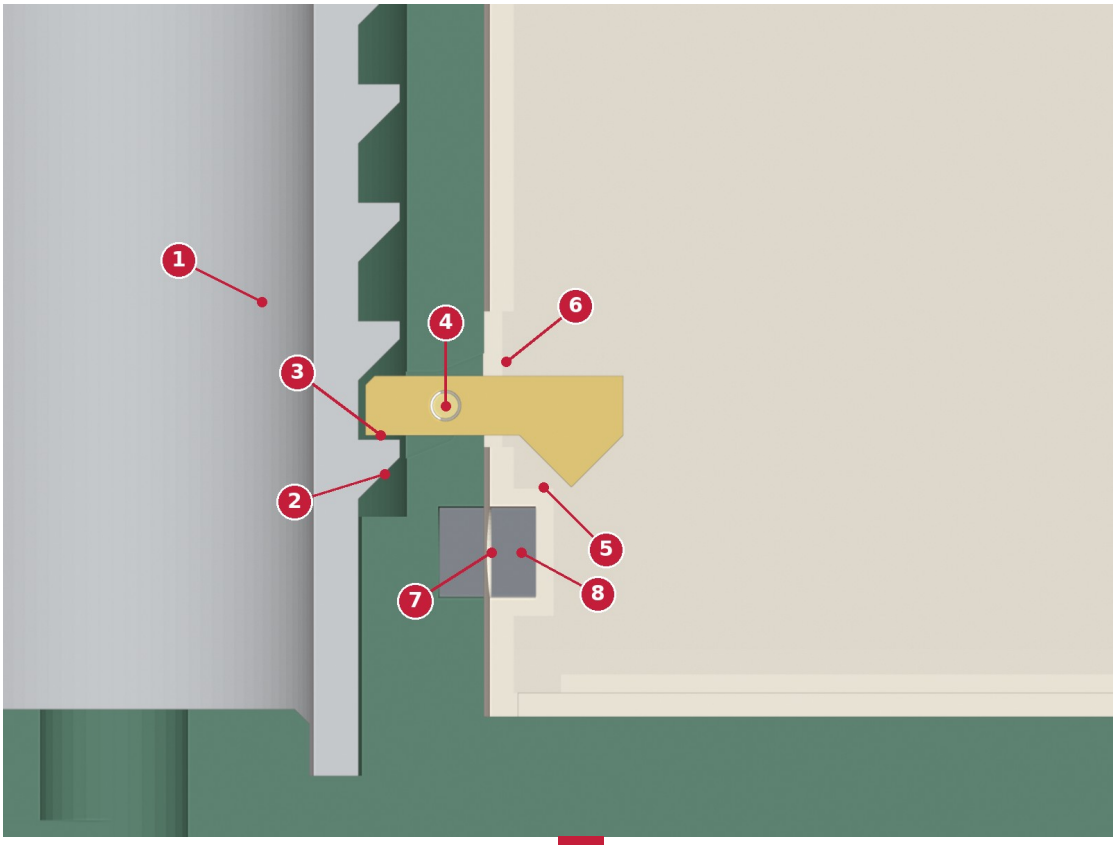
Hardware



1	Latch (printed, 24×)
2	Filament pin, Ø1.75 mm × 17.7 mm (24×) — cut from ordinary 1.75 mm filament
3	Disc magnet 6 × 3 mm for the tubes (24×)
4	Disc magnet 6 × 3 mm for the boxes (24×)
5	Self-tapping screw 2 × 6 mm (24×, optional: extra weight for the latches)
6	Filament pin, Ø1.75 mm × 9.5 mm (1×, for the star)

TIP Cut all 24 latch pins and the star pin before you start. Cut them square with flush cutters and take the burr off with a file so they slide through the holes without force.

How the mechanism works

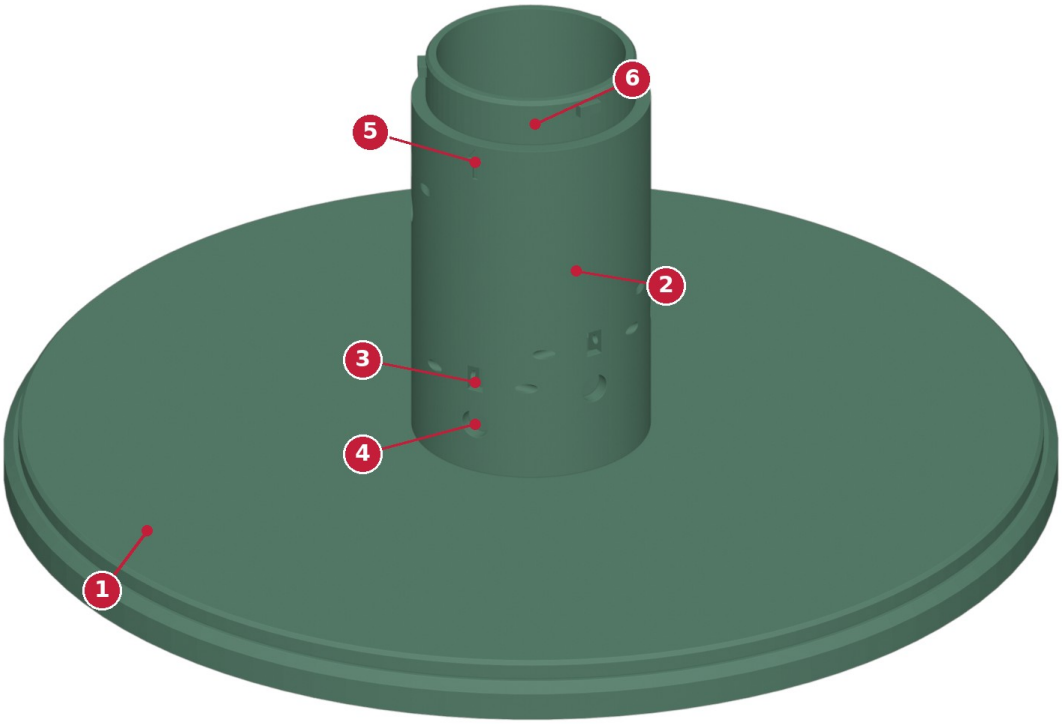


1	Drum (turned by the knob)	**5**	Tooth of the latch, hanging in the box window
2	Drum flange with one notch for this box's day	**6**	Window in the box's inner wall
3	Tail of the latch, resting just above the flange	**7**	Tube magnet
4	Pivot: filament pin glued in the tube	**8**	Box magnet

Every box has its own latch. The latch pivots on a filament pin in the tube wall. Its tooth hangs in a window in the box's inner wall; its tail rests just above a flange on the drum. When you pull a box, the tooth has to ride up its 45° face, so the tail has to go down. On every day except the right one the flange is in the way, and the box stays locked. On the box's own day the knob has turned the drum so that a notch sits under the tail. The tail drops into it and the box slides out. The two magnets pull the box back in and hold it snugly.

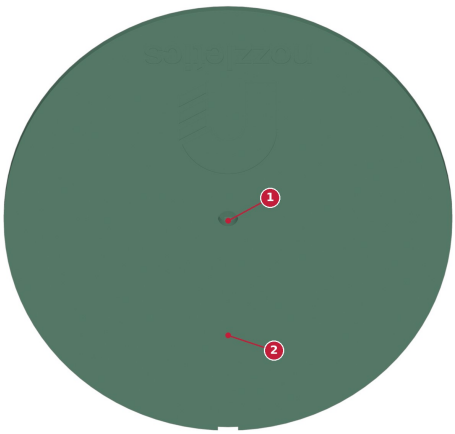
GOOD TO KNOW This is also why the boxes go in **last**, one by one, with the knob on each box's own day (Step 12). A box can only pass its latch when the notch is under the tail.

Step 1 Check the base



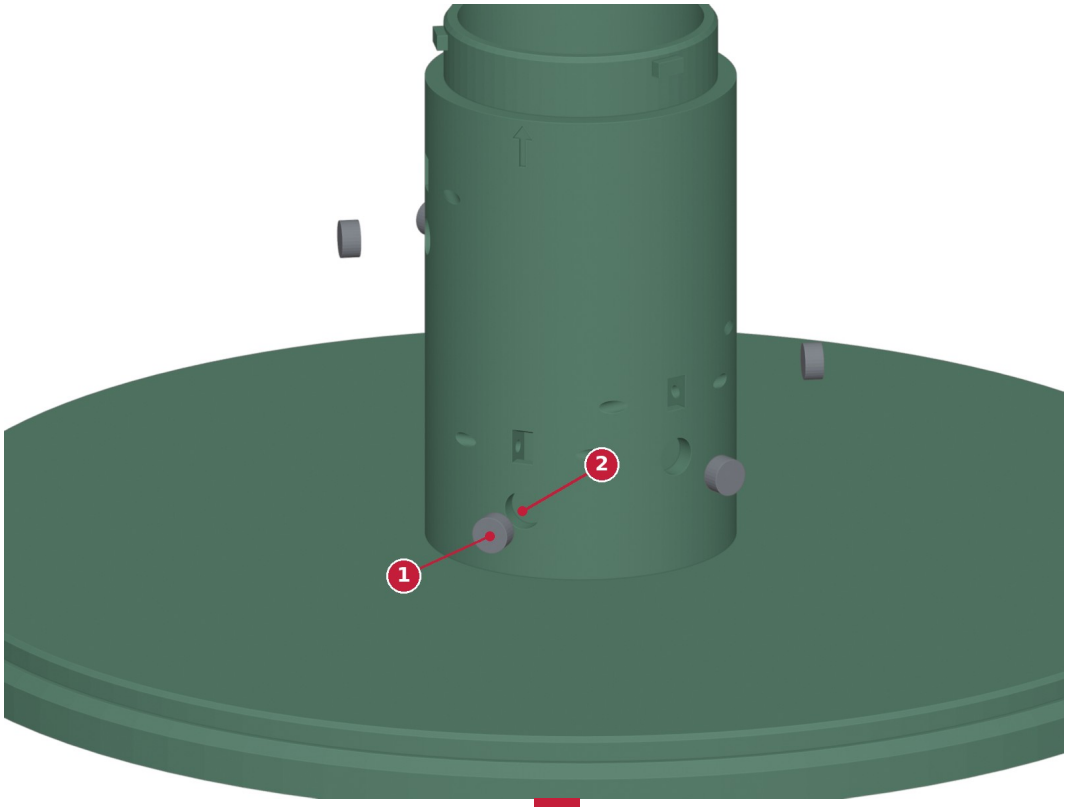
1	Base — it is also the tray of the bottom tier
2	Centre tube of tier 4
3	Latch slot (one per box, 6 per tier)
4	Magnet pocket under each slot
5	Alignment arrow under the bayonet shoulder (front of the tree)
6	Bayonet spigot for the next tier

1. Remove any stringing or elephant's foot from the slots, the pin holes beside them and the magnet pockets. A latch must swing freely in its slot.
2. Look for the small arrow just below the top shoulder of the tube. It marks the **front** of the tree. Box 19 sits here on this tier; on the tiers above, boxes 13, 7 and 1.
3. Optional garland: the base has a hole in the centre and a groove underneath that runs to the back edge (below). The cable can go up through the hollow drum.



1	Centre hole for the garland cable
2	Cable groove on the underside, out to the back edge

Step 2 Glue in the tube magnets



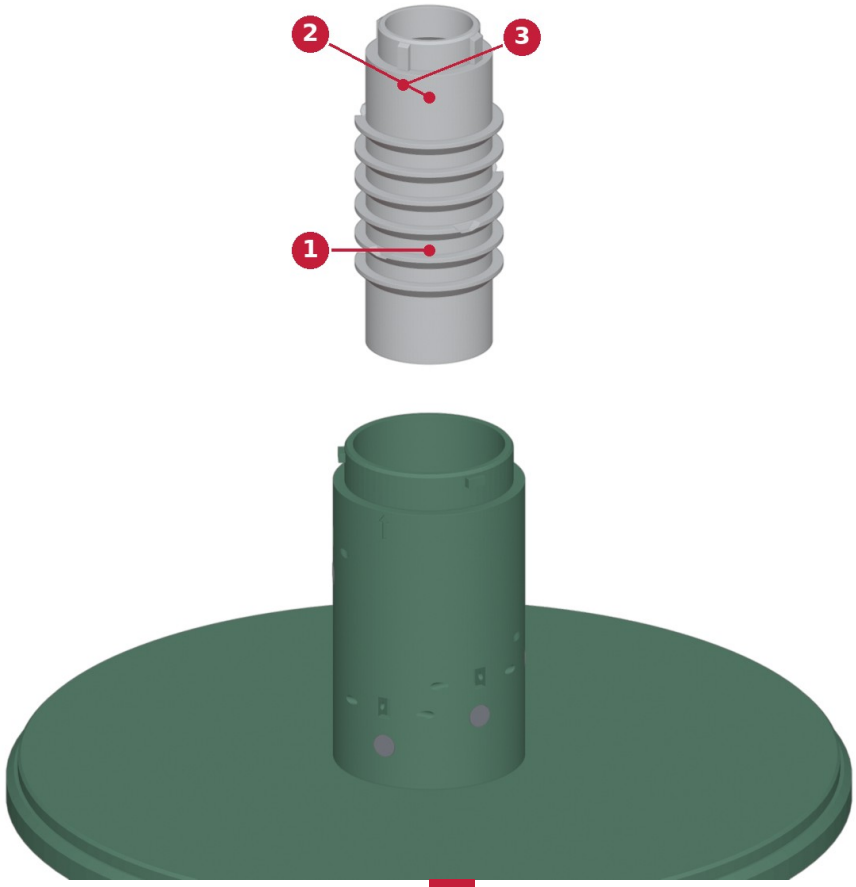
1 6 × 3 mm magnet

2 Pocket in the tube wall, open to the outside

1. Stack all 48 magnets into one column. Mark the top face of the column with a felt pen. Every magnet now shows its pole the same way. Take the magnets off one at a time.
2. Put a small drop of super glue into a tube pocket. Press a magnet in with the **marked face pointing outwards**. It must sit flush with the tube surface or a hair below it, never proud.
3. Fill all 6 pockets of this tube the same way. Wipe off excess glue and let it cure.

CAUTION All 24 tube magnets have the same pole facing outwards. The box magnets (Step 11) go in the other way round, so every box is **attracted**, never pushed away. Test with a loose magnet before the glue sets.

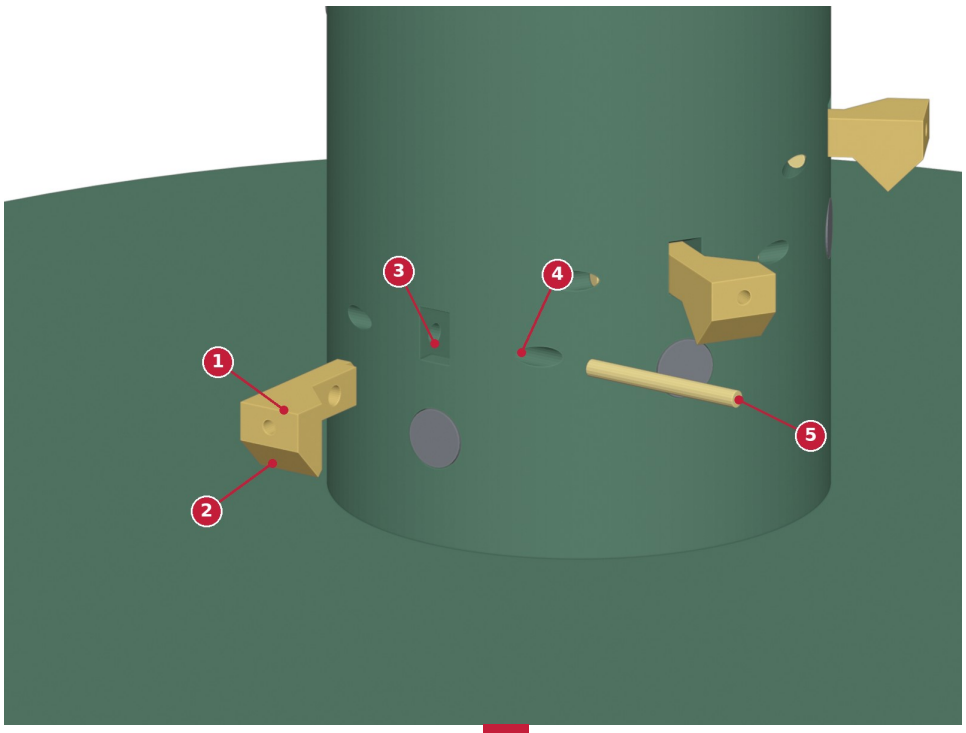
Step 3 Insert the bottom drum



1	Drum segment of tier 4	3	Wide coupling tooth
2	Coupling collar on top: it carries the next drum segment		

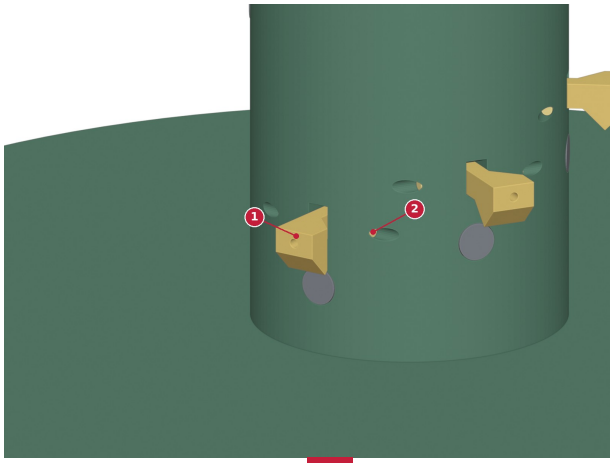
1. Lower the tier-4 drum into the tube from the top, collar up. It stands on the centring pin in the base.
2. Turn it by hand: it must spin freely with no rubbing.
3. The drum must go in **before** the latches. Once the latches are fitted, their tails sit over the drum flanges.

Step 4 Fit the latches with filament pins



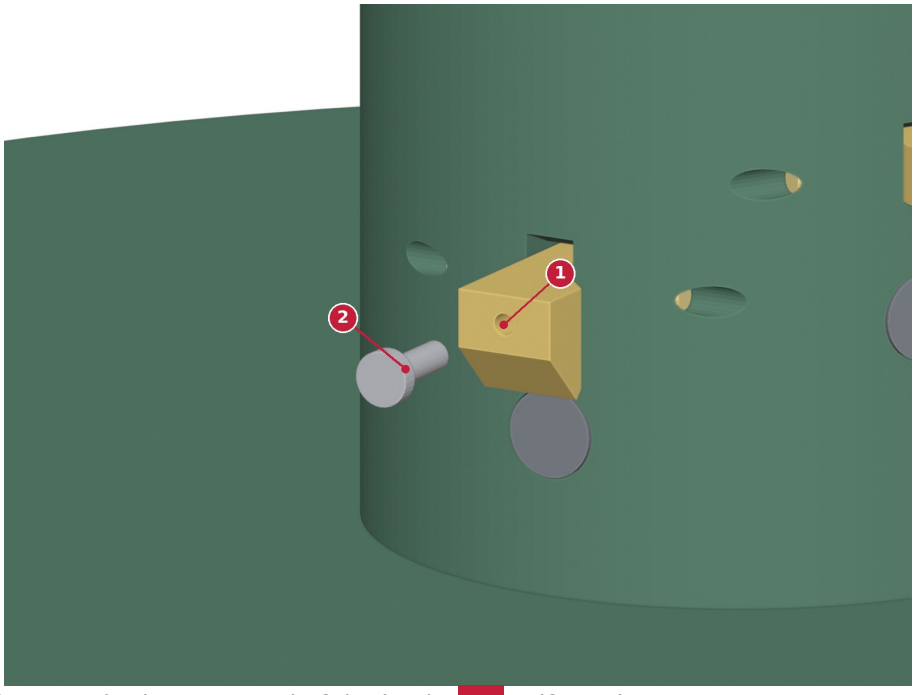
1	Latch, tail first
2	Tooth: it points **down**, outside the tube
3	Latch slot
4	Pin hole in the tube wall (one on each side of the slot)
5	Filament pin Ø1.75 × 17.7 mm

1. Push a latch into its slot from the outside, tail first, tooth hanging down. The narrow tail goes in over the drum flange.
2. Line up the hole in the latch with the two pin holes in the tube wall. Slide a filament pin through tube, latch and tube.
3. Put a tiny drop of super glue on each end of the pin, where it sits in the tube wall. **Do not glue the latch**: the latch must swing freely on the pin.
4. Check: lift the outer end. The latch must fall back onto the floor of the slot by its own weight. Repeat for all 6 latches.



1	Latch fitted: resting on the slot floor, tooth down
2	Pin glued into the tube only, ends flush with the tube surface

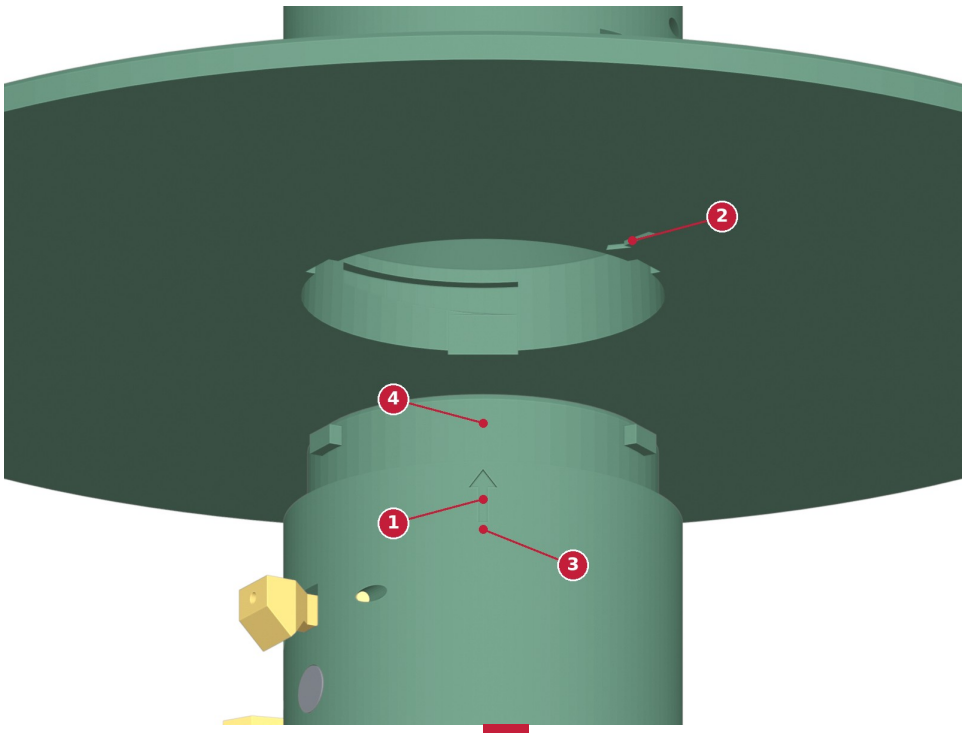
Step 5 Optional: weight screws in the latches



1 Pilot hole Ø1.5 mm in the outer end of the latch **2** Self-tapping screw 2 × 6 mm

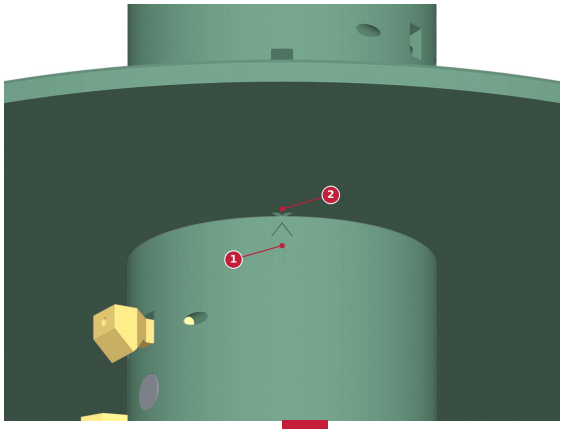
A steel screw in the outer end makes the latch fall back more reliably. The box magnet also pulls on it a little, which helps the latch fall back faster. Turn the screw in gently until the head touches. Do not over-tighten, the end of the latch is small. You can also fit the screws before Step 4.

Step 6 Mount the next tier: line up the arrows



1	Arrow on the lower tube (front of the tree)	**3**	Lugs: one is wider, so it fits only one way
2	Arrow on the underside of the upper carrier	**4**	Bayonet spigot of the lower tube

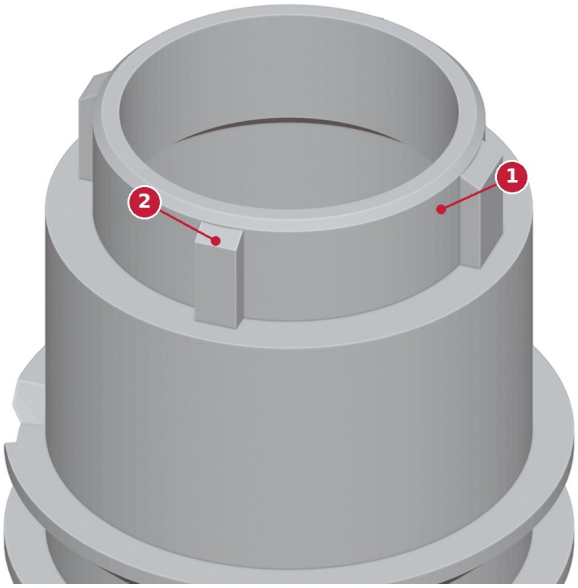
1. Glue the 6 tube magnets into the tier-3 carrier first (Step 2).
2. Hold the carrier above the tree. Turn it so its arrow is about **40° to the left** (anticlockwise) of the arrow on the lower tube. Only in this position do the three lugs, one of them wider, drop into the entry slots.
3. Lower the carrier fully onto the shoulder.
4. Turn the carrier **clockwise** (seen from above) until it stops. The ramps pull it down tight. The two arrows must now point at each other (below).



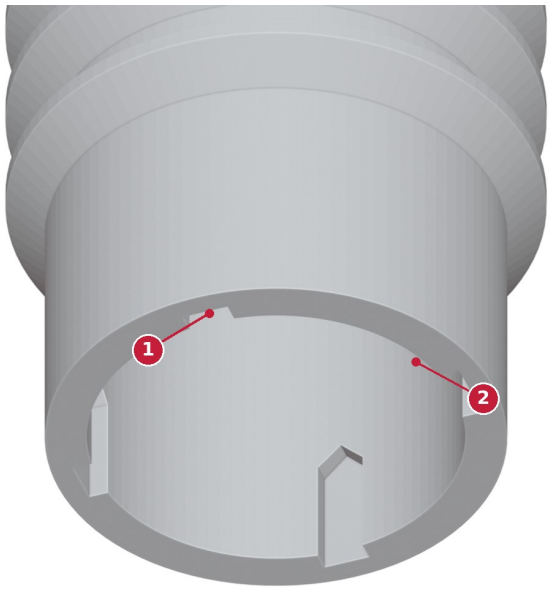
1	Lower arrow	**2**	Upper arrow: lined up = locked, and the boxes of both tiers face the same way

CAUTION If the arrows do not line up, the tier is not fully turned. All 24 days depend on this. Never force the carrier past the stop.

Step 7 Add the next drum segment



1 Collar on top of the lower drum, 4 teeth
2 The wide tooth

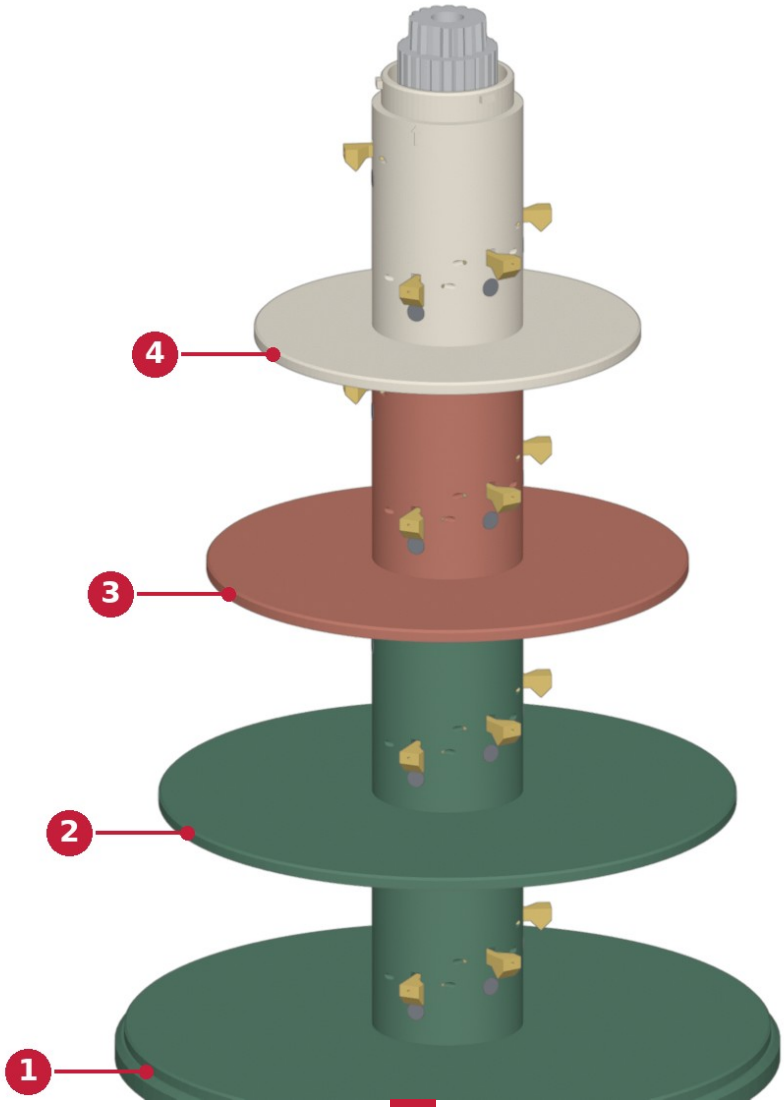


3 Wide slot in the foot of the upper drum
4 Bore of the upper drum

1. Drop the tier-3 drum into the tier-3 tube from the top.
2. Turn it slowly until the wide tooth of the collar below drops into the wide slot. The segment then sinks down about 8 mm and rests on its seat in the tube.
3. Turn the drum: both segments now turn together, and the whole drum turns freely.

GOOD TO KNOW The wide tooth makes sure every drum segment can only go in one way, so the notches for all 24 days line up with their boxes.

Step 8 Repeat for tiers 3, 2 and 1



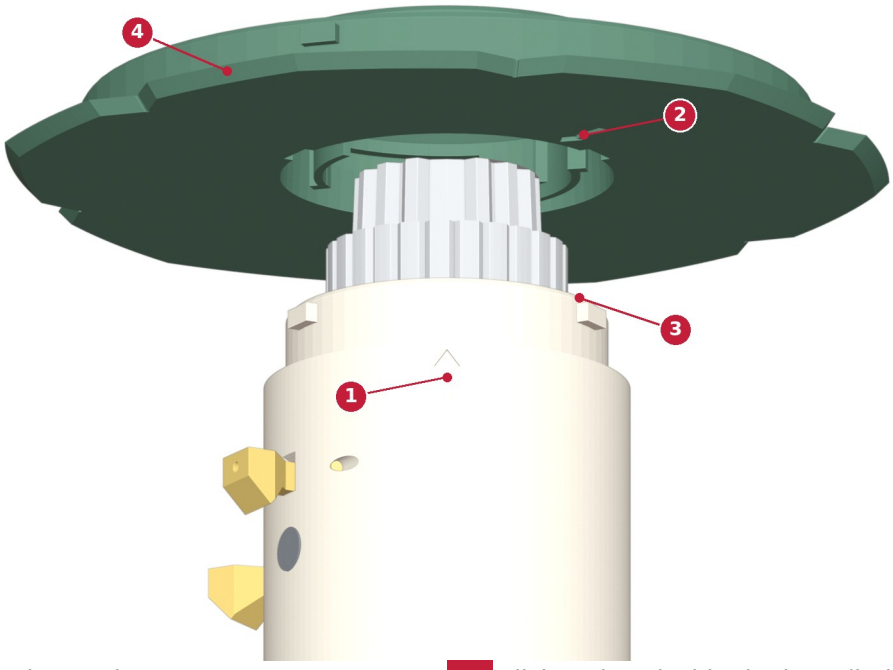
1	Tier 4 (base)	**3**	Tier 2
2	Tier 3	**4**	Tier 1 — the drum on top ends in the splined stud for the knob

For each tier, from the bottom up:

1. Tube magnets into the carrier (Step 2).
2. Carrier onto the tier below, arrows lined up (Step 6).
3. Drum segment into the tube, wide tooth into the wide slot (Step 7).
4. Latches with filament pins (Step 4), optional screws (Step 5).

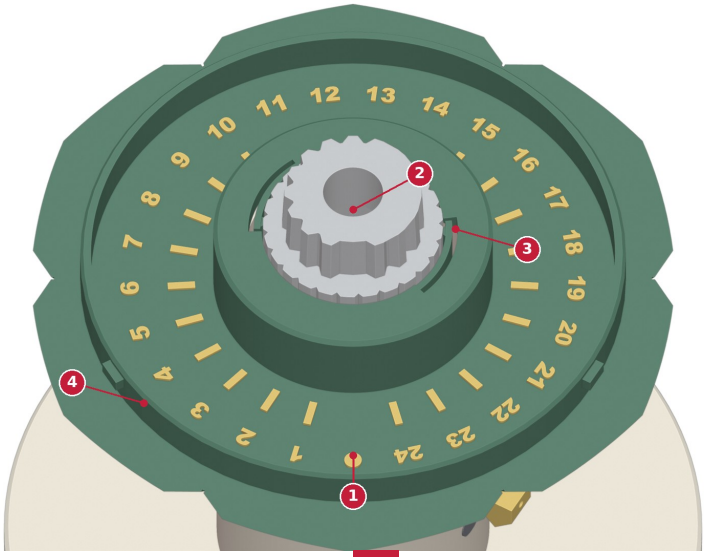
GOOD TO KNOW Leave the boxes off for now. They go in at the very end (Step 12).

Step 9 Mount the top floor



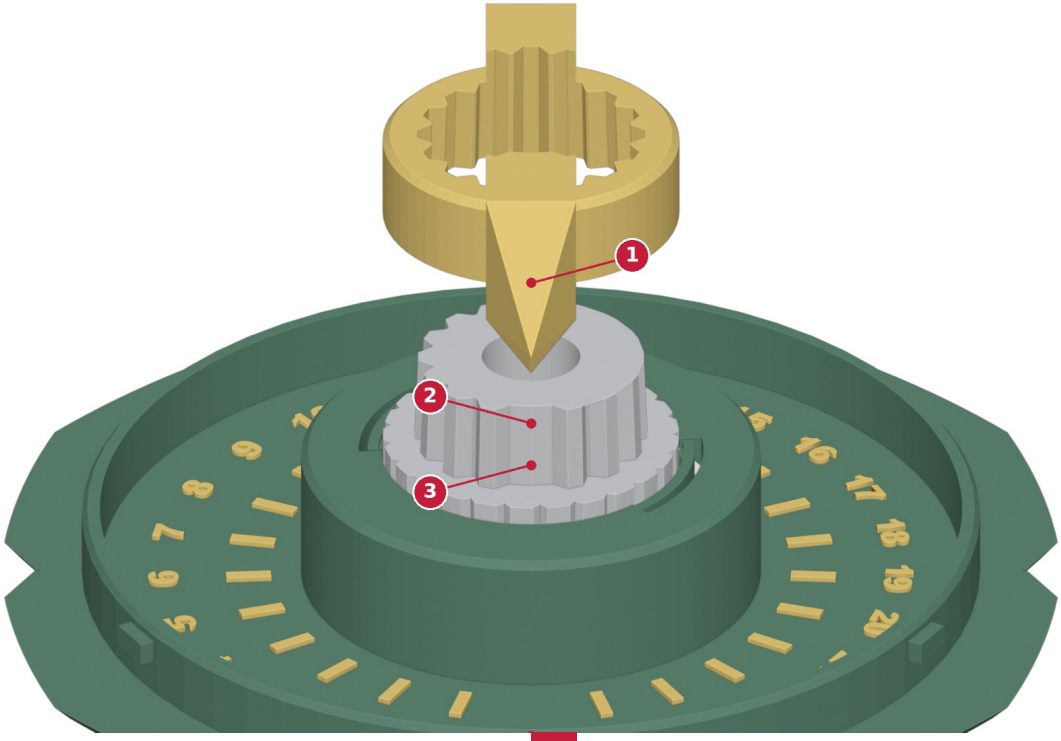
1	Arrow on the tier-1 tube	**3**	Click springs inside the boss (built in)
2	Arrow on the underside of the top floor	**4**	Top floor

1. Slide the top floor down over the top drum. The drum passes up through the boss in the middle.
2. Hold it with its arrow about 40° to the left of the arrow on the tube, lower it fully, then turn it **clockwise** until the arrows line up, exactly like Step 6.
3. Turn the drum by the stud: you should feel and hear a click at every position. The two springs in the boss drop into the grooves of the drum.



1	Dial: start dot and the days 1–24, clockwise	**3**	Click springs
2	Splined stud of the top drum	**4**	Rim with the bayonet lugs for the cap

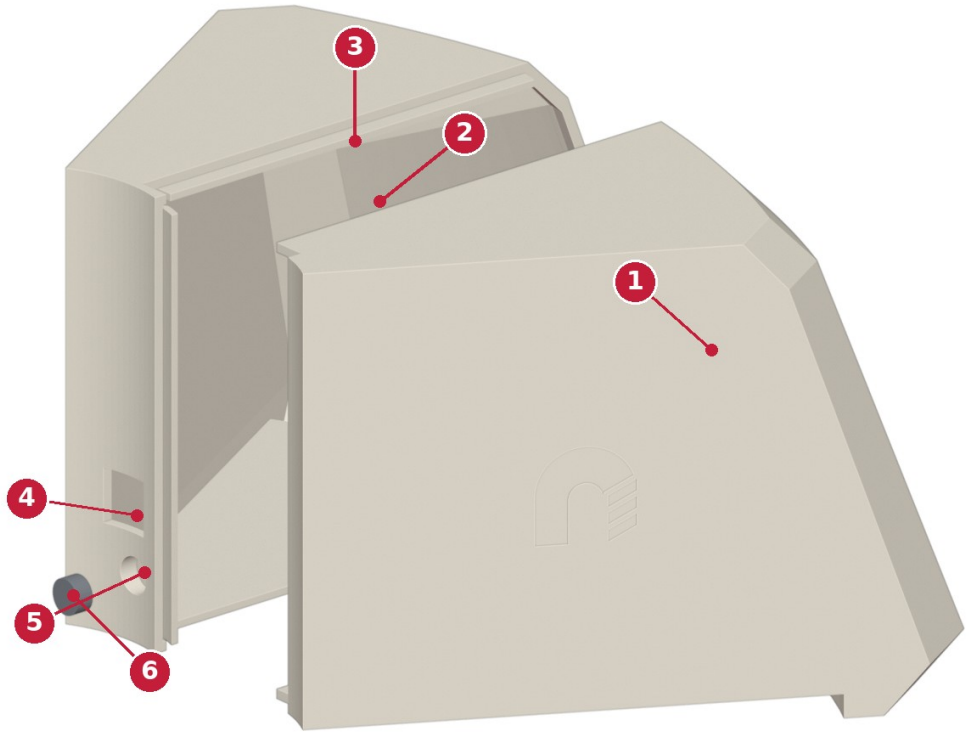
Step 10 Fit the knob



1	Knob, pointer forward	3	Splined stud
2	One tooth of the spline is missing: it belongs under the pointer	4	Start dot on the dial

1. Turn the knob until the solid part of its hole lines up with the gap in the spline. The knob fits only this way.
2. Push the knob down. The stud is slightly tapered, so it wedges tight just above the drum. Do not hammer it.
3. Turn the knob clockwise to the **start dot**. It clicks at every number.

Step 11 Prepare the boxes

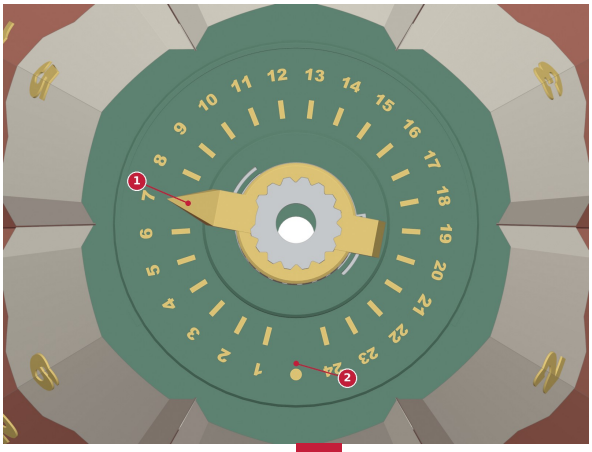


1	Half A	4	Window for the latch tooth
2	Half B	5	Magnet pocket in the inner wall of half B
3	Lip of half B: it plugs into half A	6	Box magnet 6 × 3 mm

1. Glue a magnet into the pocket in the inner wall of half B. The **marked face points inwards**, the opposite of the tube magnets, so the box is attracted. Flush or a hair below the surface.
2. Check with a spare marked magnet: the box must be pulled toward the tube, not pushed away.
3. Put the surprise in and press half B onto half A. The lip holds the halves together, so no glue. To open the box later, just pull the halves apart.

> **TIP** Number the boxes as you go. Each number has exactly one place on the tree.

Step 12 Load the boxes, one day at a time



1 Knob pointing at a day (here: 7)

2 Start dot



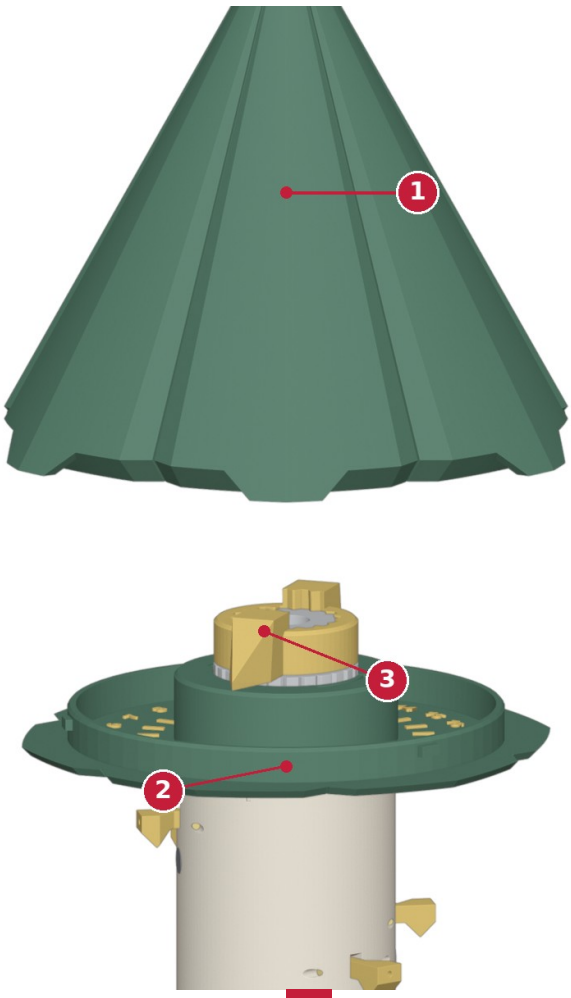
1 Box 1 slides in toward the tube

2 Its latch lifts and drops into the window, the magnets snap together

1. Turn the knob to **1**. Put box 1 onto its tray at the front of the top tier, below the arrow, and push it straight in toward the tube. The latch tooth rides up, drops into the box window, and the magnets pull the box home.
2. Turn the knob to 2 and fit box 2, then 3, and so on up to 24. Boxes 1–6 go on the top tier, 7–12, 13–18 and 19–24 on the tiers below. On every tier the lowest number sits at the front, and the numbers run to the right (anticlockwise seen from above).
3. When all 24 boxes are in, turn the knob back to the **start dot**.

CAUTION A box that will not go in is not on its day yet. Do not force it. Check the knob position and the box number.

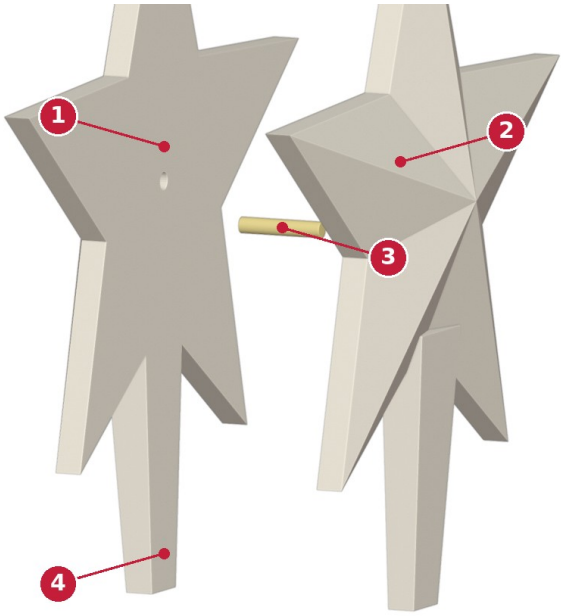
Step 13 Put on the cap



1 Cap: the space inside is a secret compartment
2 Rim lugs on the top floor (one wide lug)
3 Knob, hidden under the cap

1. Set the cap on the rim, about 40° to the **right** (clockwise) of its final position, so the lugs drop into the entry slots.
2. Turn the cap **anticlockwise** until it stops. The cap turns the opposite way to the tubes, so opening the cap never loosens the tree.
3. To remove the cap, turn it clockwise and lift it off.

Step 14 Assemble the star





1	Front half
2	Back half
3	Filament pin Ø1.75 × 9.5 mm: it centres the halves
4	Square tapered stem
5	Socket in the tip of the cap

1. Glue the short filament pin into the centre hole of one half.
2. Put glue on the flat back. Press the two halves together on the pin, star points aligned. Let it cure.
3. Push the stem into the socket at the tip of the cap. The taper wedges it tight.

Every day in December



1. Take the cap off: turn it clockwise and lift it.
2. Turn the knob clockwise **one click**, to today's number.
3. Put the cap back on (anticlockwise to lock).
4. Pull today's box straight out by its tab. Only this box comes out. Pull the halves apart to open it.
5. To put an empty box back, turn the knob to its number and push it in, as in Step 12.

Troubleshooting

Problem	Check
A box will not come out on its day	Is the knob exactly on the number (clicked in)? Is the latch stuck? Lift its outer end once: it must fall freely.
Any box can be pulled out	That latch is not falling back: glue on the latch, or a burr in its slot. Free it, or fit the weight screw.
The knob turns but nothing changes	A drum segment is not engaged: the wide tooth is not in its slot (Step 7).
Boxes of a tier point the wrong way	That tier is not locked: its arrows are not lined up (Step 6).
A box falls out or will not stay in	Magnet polarity reversed in the box or the tube (Steps 2 and 11).